Wild Rift Player Churn Prediction: Model Selection and Deployment Design

Executive Summary

Player churn is a major challenge in mobile gaming, especially in games where many users leave within their first few sessions. This project evaluates a churn prediction approach for Wild Rift with a focus on Southeast Asian markets, where cold-start behavior, uneven telemetry quality, and regional diversity make prediction harder. Three machine learning frameworks were compared: Random Forest, XGBoost, and LightGBM.

LightGBM is recommended as the final model because it offers the best balance of recall, training speed, and deployment practicality. XGBoost produced the strongest overall AUC, but LightGBM performed better on recall, which matters more in this use case because missed churners are more costly than false alarms. The project also identifies non-model risks, including missing telemetry, time-zone distortion, cold-start sparsity, and regional underrepresentation. To make the model practical in production, the recommended deployment stack combines LightGBM, SHAP-based explainability, Looker dashboards, and Google Cloud Storage.

The business value of the system comes from earlier identification of at-risk players and more targeted retention actions. Based on the scenario assumptions used in this project, the design could support meaningful recovery in monthly player lifetime value. That estimate should be treated as directional rather than guaranteed, because actual impact depends on intervention quality, production telemetry, and live player behavior.

1. Introduction

Player churn is a persistent problem in mobile gaming because user acquisition costs are high and early retention windows are short. Industry research shows that a large share of mobile players disengage very early, which makes first-week prediction especially important for live-service games (Sensor Tower, 2023; Runge et al., 2014). In a title such as Wild Rift, failing to identify at-risk players early can reduce long-term value and weaken the effectiveness of retention campaigns.

This project evaluates a churn prediction pipeline for Wild Rift with a focus on Southeast Asian markets. The analysis compares three machine learning frameworks, Random Forest, XGBoost, and LightGBM, while also examining data quality issues, deployment requirements, and explainability considerations. The goal is not simply to choose the highest-scoring model. It is to identify a model and deployment design that balance recall, speed, interpretability, and operational scalability in a realistic gaming setting.

2. Business Problem and Objectives

In churn prediction, the most important technical trade-off is usually not overall accuracy. It is the balance between false negatives and false positives. In this context, a false negative means a player is predicted to stay but actually leaves. That is often more damaging than a false positive, because the missed player receives no retention intervention at all (Perianez et al., 2016; Zheng et al., 2020).

This issue becomes more difficult in Wild Rift because of three practical conditions:

- **Cold-start behavior :** many players generate very little data during their first few sessions.
- **Regional variation :** player behavior, session timing, and engagement patterns can vary across SEA markets.
- **Imbalanced classes :** churners and retained users are not evenly distributed, which can distort standard model evaluation.

As a result, the project prioritizes recall, deployment speed, and robustness to sparse features over raw accuracy alone.

3. Model Comparison

Three models were evaluated for their suitability in this use case: Random Forest, XGBoost, and LightGBM.

3.1 Random Forest

Random Forest is useful as an interpretable baseline. It works well for early experimentation and feature exploration, and it can be paired easily with post-hoc interpretation methods. However, it performed less well on recall and was weaker at handling sparse early-session behavior. In practical terms, this means it was more likely to miss true churners, especially in underrepresented segments.

3.2 XGBoost

XGBoost is a strong gradient-boosting framework with solid predictive performance and good handling of noisy or incomplete data (Chen & Guestrin, 2016). In this project, it produced the best AUC and precision, which makes it attractive when the main goal is to reduce unnecessary intervention. However, it showed a greater tendency to overfit certain user groups unless tuned carefully. It is a strong benchmark, but it was not the best operational fit for this specific churn objective.

3.3 LightGBM

LightGBM offered the best balance between performance and deployment practicality. It trained quickly, handled large feature sets efficiently, and delivered the strongest recall among the three models (Ke et al., 2017). That matters because the business cost of missed churners is high. LightGBM also handled sparse early-session data more effectively than the alternatives, which made it better suited to cold-start conditions. Its lower native interpretability was addressed by using SHAP for post-hoc explanation (Lundberg & Lee, 2017).

3.4 Comparison Summary

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Framework	AUC	Recall	Training Speed	Cold-Start Suitability	Main Strength	Main Limitation
Random Forest	0.79	0.73	Low	Low	Transparent baseline	Higher false negatives
XGBoost	0.85	0.79	Medium	High	Strong predictive accuracy	Greater tuning complexity
LightGBM	0.84	0.82	High	Strong	Best balance of recall and speed	Lower native interpretability

Based on these results, LightGBM was selected as the final framework.

4. Evaluation Approach and Results

The evaluation used 10-fold stratified cross-validation on the Kharoua (2023) online gaming behavior dataset. The aim was to judge the models not only by overall fit, but by how well they support early identification of churn risk.

4.1 Evaluation Metrics

Metric	Definition	Why It Matters
Accuracy	Overall proportion of correct predictions	Useful summary, but less informative under class imbalance
Precision	Share of predicted churners who actually churned	Reduces wasted retention effort
Recall	Share of actual churners correctly identified	Most important for minimizing missed at-risk players
F1-score	Harmonic mean of precision and recall	Balances detection quality across both error types
False Positives	Predicted churn, but player would have stayed	Wastes retention resources
False Negatives	Predicted to stay, but player churns	Misses the chance to intervene

4.2 Key Findings

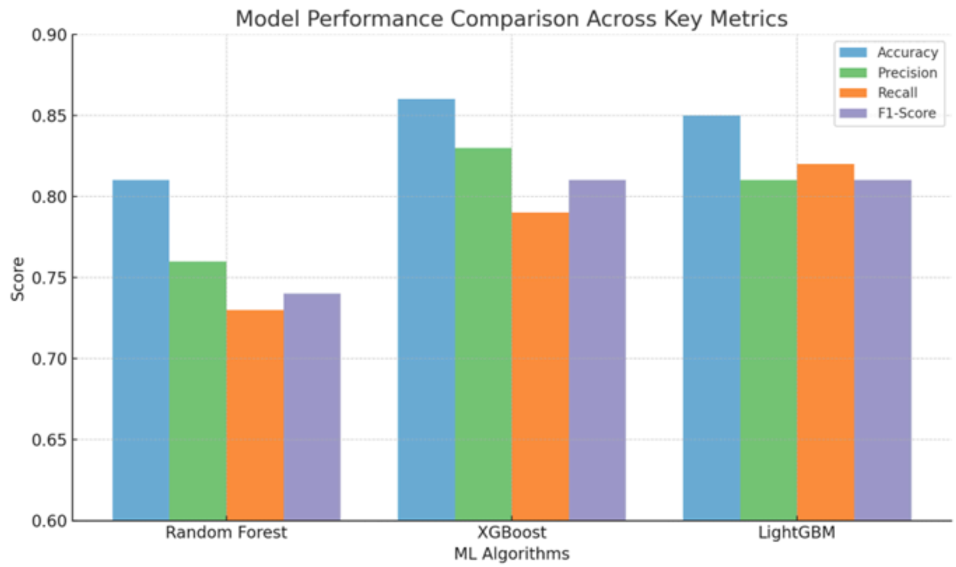
The results show a clear trade-off across the three models.

XGBoost produced the strongest AUC and precision, making it attractive for high-confidence targeting.

LightGBM produced the highest recall, which is more valuable in this use case because missed churners are costly.

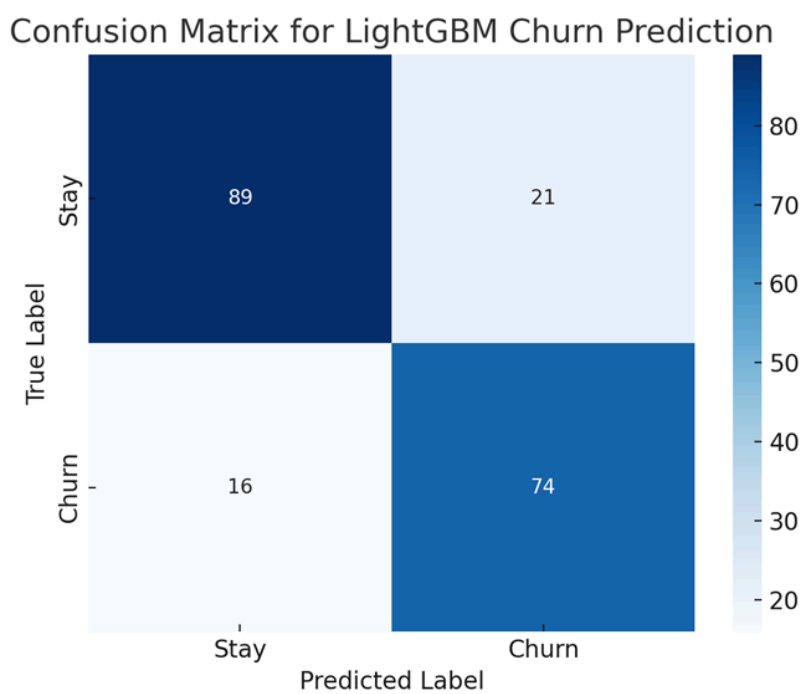
Random Forest lagged behind on recall and was therefore less suitable for production use despite its interpretability.

Figure 1. Model Performance Comparison Across Key Metrics



Comparison of Random Forest, XGBoost, and LightGBM across accuracy, precision, recall, and F1-score. LightGBM delivers the strongest recall, while XGBoost leads on AUC and precision.

Figure 2. Confusion Matrix for LightGBM Churn Prediction



The confusion matrix shows that LightGBM maintains strong recall while keeping false positives at a manageable level. This makes it practical for early intervention rather than purely retrospective classification.

5. Data Challenges and Risk Mitigation

Model choice alone does not solve the churn problem. Prediction quality also depends heavily on data quality and feature coverage.

5.1 Missing and Noisy Data

Several gameplay variables, such as average session length, purchase behavior, and win-loss ratio, contained missing or unevenly distributed values. Missingness was especially common among low-engagement users, which is problematic because those users are often the ones most at risk of churn.

To reduce this risk, the project uses median imputation, missing-value flags, and model families that tolerate incomplete feature inputs well. Gradient-boosting methods were preferred partly for this reason.

5.2 Time-Zone Distortion

The dataset records timestamps in UTC without full regional context. That can distort engagement patterns, especially in regions where session timing is culturally or behaviorally distinct.

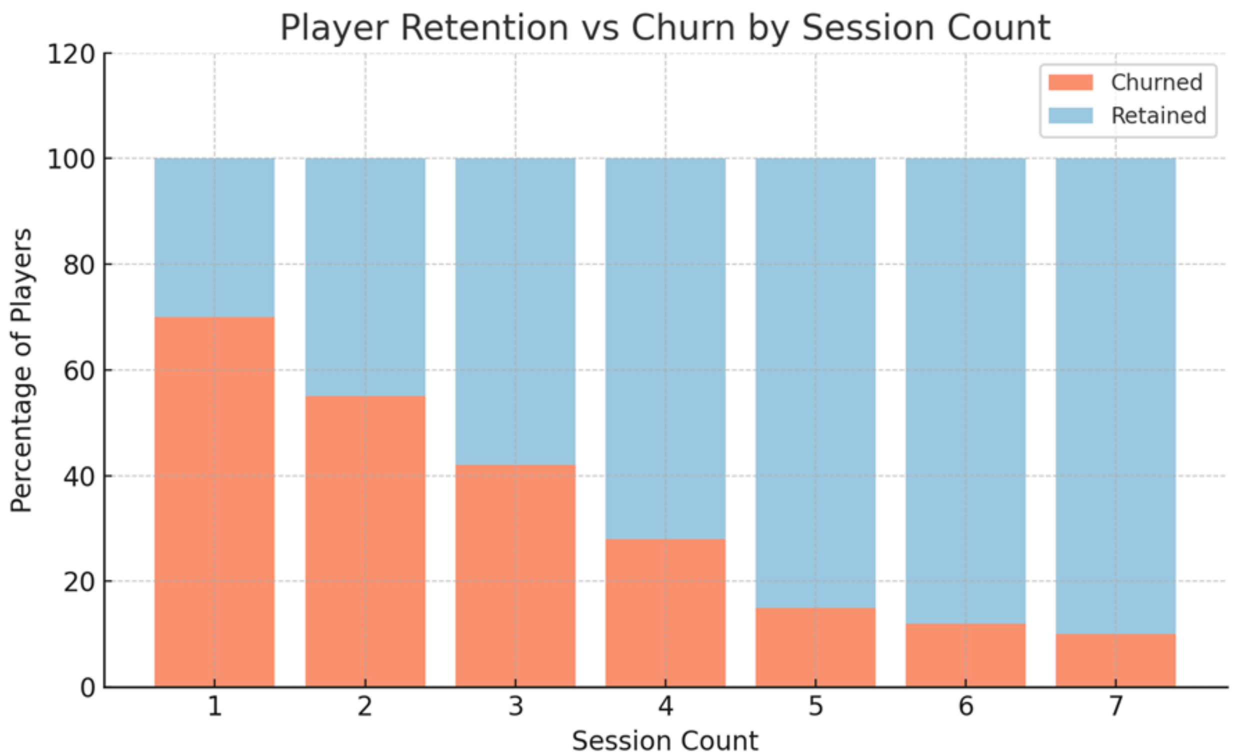
The mitigation approach is to engineer cyclical time features and approximate local behavior using region-linked metadata where available. This does not fully solve the issue, but it reduces the chance of misleading session-timing patterns.

5.3 Cold-Start Sparsity

A meaningful share of players in the dataset had very limited activity, with about 23% showing fewer than two sessions. This creates a classic cold-start problem because traditional behavioral features become too sparse to support stable classification.

To address this, the design emphasizes early-session proxies such as tutorial completion, early quit signals, and initial session depth. LightGBM was preferred in part because it handles sparse structured inputs efficiently.

Figure 3. Player Retention vs. Churn by Session Count



This figure shows that churn is concentrated in players with fewer than five sessions, which reinforces the need for early-session prediction and intervention.

5.4 Regional Underrepresentation

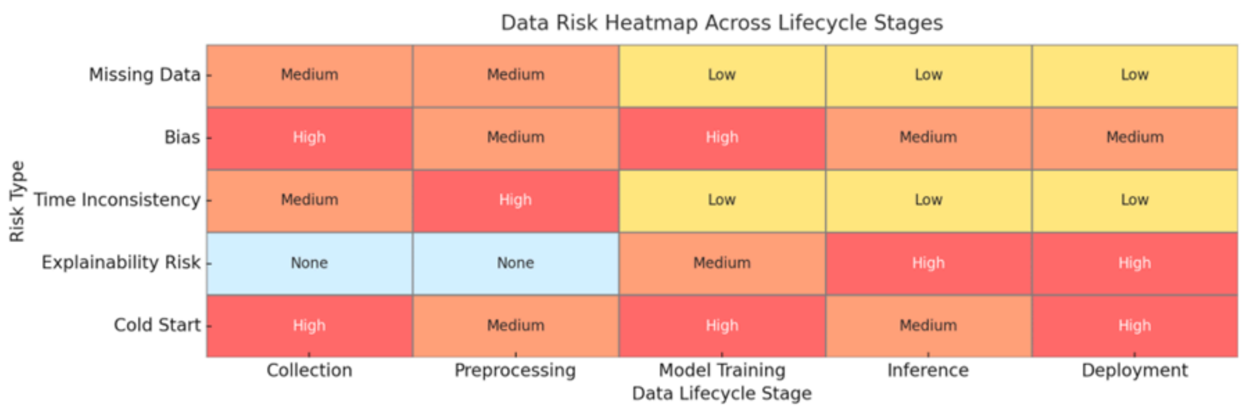
Regional imbalance can reduce model fairness and weaken performance on smaller or behaviorally distinct cohorts. In a SEA-focused deployment, this is important because player behavior can vary substantially across markets.

The proposed mitigation is to use stratified reweighting, synthetic oversampling where appropriate, and segment-level monitoring after deployment. Fairness should be treated as an ongoing validation problem rather than a one-time preprocessing fix (Google Cloud, 2023; Zhu, 2023).

5.5 Lifecycle Risk Summary

Lifecycle Stage	Problem	Risk	Resolution
Preprocessing	Missing data, UTC timestamps	Cultural misalignment, skewed patterns	Local time approximation, missing-value flags
Training	Class imbalance, overfitting	Bias toward dominant player groups	Weighted loss functions, rebalancing
Inference	Limited cold-start data	Low prediction reliability for new users	Early-session proxy features and LightGBM
Deployment	Static churn thresholds	Feedback drift, unfair targeting	Periodic refresh cycles and monitoring

Figure 4. Data Risk Heatmap Across Lifecycle Stages



The heatmap summarizes how data-quality, bias, cold-start, and explainability risks vary across the machine learning lifecycle, helping highlight where mitigation is most needed.

6. Deployment Design

The production recommendation is based on practical deployment constraints, not only model scores.

6.1 Model Layer

LightGBM is the recommended production model because it supports fast training and efficient inference while maintaining strong recall. This makes it suitable for near real-time churn scoring.

6.2 Explainability Layer

Because LightGBM is not inherently transparent, SHAP is recommended for feature attribution and post-hoc explanation (Lundberg & Lee, 2017). This helps non-technical stakeholders understand why a player is flagged as at risk and supports more accountable use of predictions.

6.3 Monitoring Layer

Looker is the preferred monitoring tool because it supports live dashboarding and can expose prediction outputs by segment, region, and feature attribution. That makes it useful for product, analytics, and operations teams that need current visibility rather than static reporting.

6.4 Infrastructure Layer

Google Cloud Storage is the preferred storage layer because it fits naturally with a cloud-native analytics stack and supports low-latency regional deployment. It is also compatible with a Vertex AI-style pipeline if the project later moves toward managed training and deployment.

6.5 Integrated Recommendation

Component	Selected Option	Reason
ML Framework	LightGBM	Best balance of recall, speed, and deployment fit
Explainability	SHAP	Makes predictions more interpretable
Monitoring	Looker	Supports real-time operational visibility
Storage	Google Cloud Storage	Strong cloud integration and regional deployment fit

Together, this stack offers a practical balance between model quality, explainability, and operational usability.

7. Business Impact and Strategic Fit

The main business value of the project comes from earlier and more targeted retention action. If at-risk players can be identified during their first few sessions, the game team can prioritize interventions such as onboarding prompts, progression nudges, incentives, or support messaging before disengagement becomes permanent.

In this project, the proposed system is associated with a scenario-based estimate of up to $1.4M in monthly LTV recovery. That figure should be treated as a directional estimate rather than a guaranteed outcome. Real impact would depend on production model quality, intervention design, player segmentation strategy, and the actual elasticity of player retention.

The project also supports broader operational goals. A more reliable churn model improves decision-making across analytics, product, and live operations teams. A more explainable model can also make it easier to justify retention targeting and monitor whether certain user groups are being over- or under-flagged.

8. Limitations

This project has several important limitations.

First, the analysis is based on an academic dataset rather than live Wild Rift production telemetry. That means deployment assumptions are illustrative rather than operationally validated. Second, some business assumptions, including projected LTV recovery, are scenario-based and should not be interpreted as guaranteed impact. Third, fairness considerations were discussed conceptually and through design choices, but not validated through a full production fairness audit. Finally, cold-start mitigation was addressed through feature engineering and model selection, but truly sparse users remain difficult to classify with high confidence.

These limitations do not reduce the usefulness of the project, but they should shape how its conclusions are interpreted.

9. Conclusion

This project compared Random Forest, XGBoost, and LightGBM for Wild Rift player churn prediction and found that LightGBM is the most practical choice for deployment. While XGBoost achieved the strongest AUC, LightGBM offered the best balance of recall, training speed, and operational fit. That makes it the better choice for a churn use case where missed churners are more costly than a moderate number of false alarms.

The analysis also shows that churn prediction is not only a modeling problem. It is a broader design problem shaped by cold-start sparsity, missing telemetry, regional imbalance, explainability needs, and deployment architecture. A useful production system therefore needs more than a strong classifier. It needs region-aware preprocessing, explainability support, monitoring, and an infrastructure stack that can handle real-time scoring responsibly.

Taken together, the project demonstrates a practical approach to churn prediction that connects model performance to business use, operational visibility, and responsible deployment.

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